Sahaj Bhadja

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Summary

Product-minded software developer with 2+ years creating web, backend, and data applications from requirements through launch. Experienced collaborating with clients, designers, engineers, and testers using TypeScript, React, Python, Java, C++, SQL, Agile delivery, and iterative feedback.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Software Engineering, Database Systems, Web Technologies, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Partnered with stakeholders to turn ambiguous workflow needs into Python APIs and Azure SQL features for 500+ users, reducing response latency from 1.2 seconds to 350 milliseconds.
- Owned improvements from design discussion through release, refactoring processing workflows and adding tests, recovery, and documentation that tripled throughput while improving maintainability, stakeholder transparency, and production reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Collaborated with product and users to develop Flask services handling 10,000+ daily requests, refining requirements and shipping authentication, grading, and feedback workflows used by more than 150 users.
- Built responsive React experiences and processing for 1,000+ files with persistence, automated tests, progress visibility, and recovery, reducing turnaround 60% while iterating continuously from user feedback.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed modular services and data integrations using Python, AWS Glue, S3, SQL, pagination, and caching, improving response times 30% and processing efficiency 68% across client-facing software applications.
- Collaborated through design discussions, code reviews, and 20+ automated test suites, accelerating bug resolution and dependable releases across Windows, Linux, and macOS while clearly documenting implementation decisions.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed JavaScript and Node.js features across eight microservices with authentication, REST APIs, WebSocket updates, and state synchronization, securely processing more than two million Azure SQL production records.
- Delivered features through functional testing, staged deployment, and monitoring with Jenkins, Docker, and Azure, reducing release time 50% while improving collaborative team feedback loops and production availability.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Worked directly with stakeholders to map complex business workflows and build React, TypeScript, and Flask features across 12+ integrations and 50,000+ records with consistent sub-200-millisecond responses.
- Presented system architecture, documented implementation choices, and resolved production issues through logging and validation, reducing manual handling 70% while supporting clear, collaborative, version-controlled release handoffs across teams.

Skills

Languages & Product: TypeScript, JavaScript, Python, Java, C++, SQL, React, Next.js, Node.js, Flask, HTML, CSS, REST APIs, WebSockets, responsive interfaces
Data & Platforms: PostgreSQL, Azure SQL, MySQL, MongoDB, AWS, Azure, GCP, Docker, Kubernetes, relational design, schema modeling, query optimization, caching
Engineering Craft: Object-oriented programming, data structures, algorithms, unit testing, integration testing, PyTest, Mockito, debugging, performance analysis, accessibility awareness, maintainable design
Consulting & Delivery: Requirements discovery, workflow mapping, client communication, design discussions, code reviews, Git, CI/CD, Jira, Agile, iterative delivery, technical documentation, production support

Projects

CPU Scheduling Simulator | Live | *C++, React, Algorithms, Interactive Visualization*
- Designed and built an interactive C++ scheduling simulator with React controls and Gantt visualization, validating edge cases and explaining performance tradeoffs across more than 1,000 process combinations.

PiXMod | Live | *React, Next.js, JavaScript, MongoDB, Stripe*
- Built a responsive full-stack product using React, Next.js, MongoDB, and Stripe, implementing maintainable user workflows, API integrations, automated validation, and subscription features that increased premium conversion 18%.

HMIS | Live | *MERN, TypeScript, REST APIs, Dashboards, Sentry*
- Built a TypeScript workflow platform with operational dashboards, REST APIs, role-based access, scheduling, notifications, and Sentry monitoring, reducing administrative workload 30% through validated data and documented releases.